

Real vs. Fake Face Detection — Comprehensive Results Report

Master 2 Thesis — Deepfake Detection Track — Phases 1-6 Complete

1. Objective

Build and validate an end-to-end pipeline for digital face forgery (deepfake) detection, then progressively evaluate and improve it following the roadmap agreed with the supervisor: (1) deeper fine-tuning, (2) cross-dataset generalization evaluation, (3) ensembling, (4) adversarial robustness. This report consolidates all five phases of the experimental work.

2. Dataset & Setup (common to Phases 1, 2 and 4)

- Dataset: "140k Real and Fake Faces" (Kaggle) — real photographs (FFHQ) vs. StyleGAN-generated faces. 100,000 train / 20,000 validation / 20,000 test images, perfectly class-balanced.
- Base architectures: ResNet18 and EfficientNet-B0, both pretrained on ImageNet.
- Training: Colab T4 GPU, Adam optimizer, batch size 64.

3. Phase 1 — Baseline (frozen backbone, linear probe)

- Only the final classification layer trained (1,026 parameters); entire ResNet18 backbone frozen at ImageNet weights.
- 5 epochs, learning rate 1e-3, ~31 minutes total training time.
- Result: 80.8% test accuracy, 0.888 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	8050	1950
True: Real	1896	8104

4. Phase 2 — Deeper Fine-Tuning (unfreeze layer4)

Rationale: Phase 1 plateaued quickly (~80% by epoch 1) because a single linear layer on frozen ImageNet features has limited capacity to specialize to forgery-specific artifacts. Phase 2 unfreezes the last residual block (layer4) of ResNet18, allowing the network's highest-level convolutional features to adapt directly to the task, while earlier layers (generic edges/textures/colors) remain frozen.

- 8,394,754 trainable parameters (vs. 1,026 in Phase 1).
- 5 epochs, learning rate 1e-4 (lowered vs. Phase 1 to protect pretrained weights given the much larger trainable capacity).
- Training accuracy rose steadily to 99.3%; validation accuracy to 98.7%, with only a small train/val gap — no meaningful overfitting on the full dataset.
- Result: 98.8% test accuracy, 0.999 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	9909	91
True: Real	146	9854

5. Phase 1 vs. Phase 2 — Comparison

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Trainable parameters	1,026	8,394,754
Test Accuracy	80.8%	98.8%
Test AUC	0.888	0.999
Precision / Recall / F1 (avg)	0.81 / 0.81 / 0.81	0.99 / 0.99 / 0.99

6. Interpretation (Phases 1-2)

- Unfreezing layer4 produced a large, consistent improvement across every metric, confirming that forgery-specific cues require feature adaptation beyond generic ImageNet representations — consistent with the CNN-based detection literature reviewed in the thesis's state-of-the-art chapter (Section 4.1).
- Both models remain well-balanced across classes (no bias toward predicting real or fake), and Phase 2's errors are small in absolute terms (237 misclassifications out of 20,000).
- Important caveat: both results above are in-distribution — same dataset and generator (StyleGAN) used for both training and testing. This is not yet a generalization result.

7. Phase 3 — Cross-Dataset Evaluation (Generalization / Drift)

Both trained models were evaluated, without any retraining, on an independent dataset built from real face-swap deepfake videos (Kaggle: "Deepfake and Real Images"), a different manipulation family from the StyleGAN full-face synthesis used for training. Test set size: 10,905 images.

Cross-dataset results:

Model	In-dist. Acc / AUC	Cross-dataset Acc / AUC
Baseline #1 (frozen backbone)	0.808 / 0.888	0.487 / 0.489
Phase 2 (fine-tuned layer4)	0.988 / 0.999	0.495 / 0.525

Confusion matrix — Baseline #1 on cross-dataset:

	Pred: Fake	Pred: Real
True: Fake	289	5203
True: Real	393	5020

Confusion matrix — Phase 2 (fine-tuned) on cross-dataset:

	Pred: Fake	Pred: Real
True: Fake	99	5393
True: Real	120	5293

Interpretation

- Both models collapse to near-chance performance on the unseen dataset (AUC 0.489 and 0.525 respectively, vs. 0.888 and 0.999 in-distribution) — a complete loss of discriminative power outside the training distribution.
- Both models are heavily biased toward predicting "Real": Baseline #1 predicts "Fake" for only 6.3% of images (682/10,905); Phase 2 for only 2.0% (219/10,905). Rather than failing randomly, both models systematically default to "Real" on this unseen data.
- Notably, Phase 2 — despite its far higher in-distribution accuracy (98.8%) — shows a slightly worse "Fake" recall on cross-dataset data than Baseline #1, suggesting that deeper fine-tuning made the model more specialized to StyleGAN-specific artifacts rather than more generally robust.
- This is consistent with the literature: StyleGAN (full-image synthesis) and face-swap deepfakes leave fundamentally different visual traces (global statistical/texture artifacts vs. localized blending-boundary artifacts, per Face X-Ray and related work, SOTA Section 4.1), so a detector trained only on one leaves no learned signal for the other.
- This constitutes the thesis's first empirical generalization/drift result, directly evidencing the cross-generator generalization gap identified as the central open challenge in Section 8 of the state-of-the-art review.

8. Phase 4 — Ensemble (ResNet18 + EfficientNet-B0)

A second, independently trained architecture (EfficientNet-B0, last feature block + classifier head fine-tuned, 414,722 trainable parameters) was combined with the Phase 2 ResNet18 via simple probability averaging, and evaluated on both the in-distribution and cross-dataset test sets used previously.

Results:

Model	In-dist Acc	In-dist AUC	Cross Acc	Cross AUC
ResNet18 (Phase 2)	98.81%	0.999	49.45%	0.525
EfficientNet-B0	93.82%	0.987	48.88%	0.499
Ensemble (avg)	98.99%	0.999	49.35%	0.498

Interpretation

- In-distribution, the ensemble matches or marginally exceeds ResNet18 alone (98.99% vs. 98.81% accuracy) — a negligible improvement, as EfficientNet-B0 alone was the weaker of the two models (93.8% accuracy).
- On the cross-dataset test, the ensemble's AUC (0.498) is slightly lower than ResNet18 alone (0.525), not higher. EfficientNet-B0's cross-dataset AUC (0.499) is essentially exact chance level, so averaging it with ResNet18's weak-but-non-random signal pulls the combined result toward chance rather than improving it.
- This demonstrates that architectural diversity alone does not resolve the generalization gap identified in Phase 3: both models were trained on the same single-source (StyleGAN) data and independently learned the same blind spot with respect to face-swap deepfakes, so ensembling combines two instances of the same failure mode rather than compensating for it.
- This result supports the conclusion that the bottleneck is training data diversity rather than model architecture or capacity — motivating mixed-source training or domain-generalization techniques as more promising directions than further ensembling.

9. Phase 5 — Adversarial Robustness

The Phase 2 fine-tuned ResNet18 was evaluated against two standard white-box adversarial attacks — FGSM (single-step) and PGD (10-step iterative) — on a 1,000-image evaluation subset (500 real / 500 fake), across a range of perturbation strengths (epsilon, in [0,1] pixel space).

Results:

Epsilon	FGSM Accuracy	PGD Accuracy
0.00 (clean, no attack)	99.0%	99.0%
0.01	0.0%	0.0%
0.03	0.1%	0.0%
0.05	13.2%	0.0%
0.10	44.9%	0.0%

Interpretation

- Clean accuracy (epsilon = 0) is 99.0%, consistent with Phase 2's in-distribution test accuracy (98.8%) — confirms the evaluation setup is correct.
- At epsilon = 0.01 — a perturbation small enough to be near-imperceptible to a human observer — accuracy collapses to 0% under both FGSM and PGD. The model is fooled on virtually every image by a single gradient-based nudge to the input pixels.
- PGD accuracy remains at exactly 0% across all tested epsilon values, including the smallest. This is expected, as PGD (iterative) is strictly at least as strong an attack as FGSM (single-step), and indicates essentially zero adversarial robustness at any perturbation strength tested.
- FGSM's accuracy is non-monotonic with epsilon (0% -> 0.1% -> 13.2% -> 44.9% as epsilon increases from 0.01 to 0.1). This is a documented property of FGSM: as a single-step linear approximation of the loss surface, it becomes a progressively worse approximation at larger epsilon, occasionally causing the perturbed sample to land back on the correct side of the decision boundary by chance. This is not observed with PGD, which remains at 0% throughout, consistent with PGD's greater reliability as an attack.
- No adversarial training or other defense was applied — this result characterizes the undefended model's baseline vulnerability, which is expected to be high for any standard CNN classifier not specifically trained for robustness, consistent with the broader adversarial ML literature.

10. Phase 6 — Mixed-Source Training

Motivated by the Phase 3 finding (generalization gap) and the Phase 4 finding (architecture/ensembling does not fix it), this phase tests whether the gap is fixable through training data diversity. A fresh ResNet18 was fine-tuned (identical recipe to Phase 2: unfreeze layer4) on a combined training set drawn from both sources used so far — StyleGAN-generated faces and face-swap deepfake video frames (30,000 real + 30,000 fake training images, 4,000 + 4,000 validation images, balanced across both sources) — then evaluated on held-out test sets from each source individually, compared directly against the single-source Phase 2 model.

Results:

Model	Test Set	Accuracy	AUC
Phase 2 (single-source)	StyleGAN	98.68%	0.999
Phase 2 (single-source)	Face-swap	49.78%	0.525
Phase 6 (mixed-source)	StyleGAN	95.49%	0.995
Phase 6 (mixed-source)	Face-swap	82.42%	0.928

Interpretation

- On the face-swap test set — the domain Phase 2 was essentially blind to (49.8% accuracy, 0.525 AUC, near chance level) — the mixed-source model achieves 82.4% accuracy and 0.928 AUC, a gain of +32.6 accuracy points and +0.404 AUC. This closes roughly 85% of the total possible gap between chance-level performance (AUC 0.5) and near-ceiling performance (AUC 1.0).
- On the StyleGAN test set, the mixed-source model shows only a small reduction relative to Phase 2 (95.5% vs. 98.7% accuracy; 0.995 vs. 0.999 AUC) — the model retains strong performance on the original domain while gaining substantial competence on the new one, rather than trading one for the other.
- This result directly confirms the hypothesis raised by the Phase 3/4 findings: the generalization failure was primarily a consequence of training data diversity, not a fundamental limitation of the CNN architecture. Where Phase 4 (ensembling two models trained on the same single-source data) failed to improve cross-domain performance, diversifying the training data itself produced a large, measurable improvement.
- This constitutes the thesis's strongest positive empirical result: a concrete, evidenced remedy for the generalization gap identified earlier in the work, achieved without any architectural changes beyond what was already used in Phase 2.

11. Overall Conclusion (Phases 1-6)

Across six phases, this work establishes a complete empirical narrative. A fine-tuned ResNet18 achieves excellent in-distribution performance (98.8% accuracy, 0.999 AUC) when trained and tested on a single generator family (StyleGAN, Phases 1-2), but this performance does not reflect a generally robust deepfake detector: the model collapses to near-chance accuracy on an unseen face-swap deepfake dataset (Phase 3), and combining it with an independently trained EfficientNet-B0 via ensembling does not recover this generalization loss, since both models share the same single-source blind spot (Phase 4). The model additionally exhibits near-total vulnerability to imperceptible adversarial perturbations (Phase 5). However, when the training data itself is diversified to include both forgery types (Phase 6), the model recovers the large majority of its lost generalization performance (Face-swap AUC: 0.525 -> 0.928) while retaining strong performance on the original domain. Together, these results provide direct empirical evidence for the central open challenge identified in the thesis's state-of-the-art review (Section 8) — and further demonstrate that, of the remedies tested, training data diversity is substantially more effective than architectural ensembling at addressing the cross-generator generalization gap. Adversarial robustness (Phase 5) remains an open weakness not addressed by mixed-source training, and is a natural direction for further work (e.g. adversarial training).

12. Next Steps

- Adversarial training (or other defense, e.g. input preprocessing/randomization) to address the Phase 5 robustness gap, which mixed-source training (Phase 6) did not target.
- Extend mixed-source training to additional forgery families (e.g. diffusion-generated faces) to test whether the Phase 6 result generalizes further, or whether it is specific to the two sources tested.
- Consider frequency-domain features (SOTA Section 4.2) or a foundation-model-based approach (e.g. CLIP adapter, SOTA Section 4.4) as potential further improvements to both generalization and robustness.

Code & results (Colab notebooks, outputs included): [\[GitHub repository link\]](#)